

AIE 2102	ESSENTIALS OF ARTIFICIAL INTELLIGENCE	L	T	P	C
SDG: 8		3	0	0	3

COURSE OBJECTIVES:

- COB1:** To familiarize with the concepts of Artificial Intelligence.
- COB2:** To learn the different searching techniques in AI.
- COB3:** To understand the fundamentals of knowledge presentations and reasoning.
- COB4:** To gain knowledge on planning and learning in artificial neural networks.
- COB5:** To explore the stages and knowledge acquisition in expert systems.

MODULE I INTRODUCTION TO ARTIFICIAL INTELLIGENCE L:9 T:0 P:0

Introduction - AI Problems and techniques - Intelligent agents – Nature of environments - Concept of rationality - Agent architectures - AI programming languages – LISP and PROLOG - Problem solving – Game playing – Minimax algorithm – Alpha beta pruning.

MODULE II SEARCH TECHNIQUES L:10 T:0 P:0

Blind search strategies - Breadth first search - Depth first search – Iterative deepening search – Uniform cost search - Heuristic search techniques - Hill climbing - Best first – Greedy search - A* algorithm - AO* algorithm – Constraint satisfaction problem.

MODULE III KNOWLEDGE REPRESENTATION AND REASONING L:9 T:0 P:0

Knowledge representation – Propositional logic – First - Order predicate logic – Semantics and interpretations – Unification – Inference – Forward and backward chaining – Rule - Based systems – Semantic networks – Frames – Ontologies – Temporal and spatial knowledge representation.

MODULE IV PLANNING AND LEARNING L:8 T:0 P:0

Planning – Types – Tools and techniques – Planning under uncertainty – Markov Decision Processes - Temporal and continuous planning – Real-time planning – Fundamentals of ANN - Biological neurons – Perceptron – Multilayer perceptron – Backpropagation.

MODULE V EXPERT SYSTEM L:9 T:0 P:0

Fundamentals of expert system – Lifecycle Model - Rule – Based expert system – Knowledge Acquisition - Applications - Intrusion detection expert system – diagnosis and treatment advisory systems – Case study.

L – 45; T – 0; P – 0; Total Hours: 45

TEXT BOOKS:

1. Stuart J. Russell and Peter Norvig, "Artificial Intelligence - A Modern Approach", Prentice Hall Publishers, ISBN 978-1292401133, 2021.

2. Raymond Lee, "Natural Language Processing: A Textbook with Python Implementation", Springer, ISBN 978-981-96-3207-7, 2025.
3. N. P. Padhy, S. P. Simon and M. Senthil Kumar "Artificial Intelligence", Oxford University Press, ISBN 978-9361032752, 2025.

REFERENCES:

1. Gaurav Raj, Bhuvan Unhelker, Ankur Choudhary Abdellah Idrissi, "Advances in Artificial Intelligence and Machine Learning: Select Proceedings of ICAAAIML 2023, Volume 1", Springer, ISBN: 978-981-97-9506-2, 2025.
2. Abdellah Idrissi, "Modern Artificial Intelligence and Data Science", Springer, ISBN 978-3-031-65037-6, 2024.
3. Sandeep Kumar, Kavita Sharma, "Computational Intelligence and Machine Learning", Soft Computing Research Society, ISBN 978-81-975670-5-6, 2024.
4. Ameet V. Joshi, "Machine Learning and Artificial Intelligence" Springer, ISBN: 978-3-031-12281-1, 2023.

COURSE OUTCOMES: Students who complete this course will be able to

- CO1:** Identify the essential concepts of artificial intelligence.
- CO2:** Analyse the search techniques and solve AI problems.
- CO3:** Interpret the knowledge representation on decision making in AI.
- CO4:** Apply planning techniques under uncertainty, formulate and solve real-time planning strategies for intelligent systems.
- CO5:** Design a simple expert system for real-world domains such as intrusion detection, diagnosis, and treatment advisory systems.